

Activity - 2

1. A CCTV image contains Salt and pepper noise due to transmission error.

Technique used: Median Filter.

Reason:- Salt and pepper noise appears as random black and white pixels due to transmission error or faulty sensor.

2. A Satellite Image is corrupted by Gaussian noise from sensor.

Technique used: Gaussian Filter

Reason:- The Gaussian Filter smooths the image and effectively reduces Gaussian noise caused by imaging sensor while maintaining overall image quality.

3. A medical x-ray image has random noise, but fine edges must be preserved.

Technique used: Bilateral Filter

Reason:- The Bilateral Filter removes random noise while preserving important edges, making it suitable for medical images where fine details are essential.

4. A Scanned document has black & white dots scattered randomly.

Technique used: Median Filter.

Reason: The Median filter removes scattered black and white dots (salt and pepper noise) without affecting the text or document edges.

5. A low light image shows heavy grain without clear boundaries.

Technique used: Non-Local Means Denoising.

Reason: This technique effectively removes heavy grain noise while preserving image details and textures better than traditional filters.

6. A foggy road image has very low contrast.

Technique used: Histogram Equalization.

Reason: Histogram Equalization improves the image contrast, making roads and objects more visible in foggy conditions.

7. An underwater image appears dull and washed out.

Technique used: CLAHE (Contrast Limited Adaptive Histogram Equalization)

Reason:

CLAHE enhances local contrast and improves the visibility of underwater objects without amplifying noise excessively.

8. A chest x-ray shows poor visibility of internal structures

Technique used: CLAHE

Reason:-

CLAHE increases local contrast, making internal organs and tissues more visible while preserving important medical details.

9. A night-time traffic image lacks intensity variation.

Technique used: Histogram Equalization.

Reason:- Histogram equalization improves brightness and contrast, making vehicles and roads easier to identify in low-light conditions.

10. A grayscale image uses only a narrow intensity range.

Technique used: Contrast stretching.

Reason:- Contrast stretching expands the narrow intensity range to use the full grayscale spectrum, resulting in a clearer and higher contrast image.

